PROGRAMME PROJECT REPORT

BACHELOR OF COMPUTER APPLICATIONS (BCA)

Programme Name :- Bachelor of Computer Applications (BCA)

Eligibility :- 10+2 or equivalent

Minimum Duration :- 3 years (6 Semesters)

Maximum Duration :- 5 years

A. Programme's Mission and Objective:-

Mission

The mission of the Smt. Chandaben Mohanbhai Patel Institute of Computer Applications (CMPICA) at CHARUSAT, Changa is to provide a world-class education in the field of computer science and applications and to produce competent computer professionals with the ability to face future challenges. The institute aims to create a learning environment that fosters creativity, innovation, and critical thinking among students.

The mission of online Bachelor of Computer Applications (BCA) program is to provide high-quality education to students in the field of computer science and applications and software development, with the aim of preparing them for successful careers in the industry. The program aims to impart knowledge and skills that enable students to develop, design, and maintain software applications effectively and efficiently.

Objectives

The online Bachelor of Computer Applications (BCA) programme is a three-year undergraduate programme that focuses on computer applications and software development. The objectives of the BCA programme may include:

1. To provide a strong foundation in the fundamentals of computer science, programming, algorithms, and database management.
2. To equip students with the necessary skills and knowledge to design, develop, and maintain software applications.
3. To develop problem-solving, analytical, and critical thinking skills in students.
4. To prepare students for a career in the field of computer applications and software development.
5. To expose students to the latest trends and technologies in the field of computer science and software development.
6. To provide students with hands-on experience through practical and project-based learning.
7. To foster teamwork and communication skills in students.
8. To promote ethical and professional behaviour among students.
9. To encourage research and innovation in the field of computer science and software development.
10. To prepare students for further studies in the field of computer science, such as a Master's degree in Computer Applications or Computer Science.

B. Relevance of the program with HEIs Mission and Goals:-

The online BCA programme is aligned with CMPICA's mission to create a learning environment that fosters creativity, innovation, and critical thinking among students. The programme is designed to provide a world-class education in computer science and applications, with a focus on developing skilled professionals who can make valuable contributions to the industry and society. The BCA programme also aims to provide an inclusive and diverse learning environment, which is essential for the development of professionals who can work effectively in multicultural and global environments. The programme provides opportunities for students to engage in practical and project-based learning, which helps them develop teamwork skills, leadership skills, and communication skills. These are essential skills that prepare students to become successful professionals in the field of computer science and applications.

C. Nature of prospective target group of learners:-

The students who wish to join BCA programme must have completed 10+2 or equivalent.

D. Appropriateness of programme to be conducted in Open and Distance Learning mode to acquire specific skills and competence:-

The Open and Distance Learning mode of University system places greater emphasis on the learner, where most of the instruction is delivered through distance mode with only a minimal component of face-to-face communication. Students will have the flexibility to learn at their own pace. They can access course materials and resources online and progress through the programme as per their convenience. It allows students to attend classes from anywhere, eliminating geographical barriers and offering more opportunities for students to learn and acquire specific skills and competencies. The Open and Distance Learning mode will provide an opportunity to learn in a technology-driven environment for the students where they can access online resources and learning materials, attend virtual classrooms, participate in online discussions, and engage with other students and faculty through various collaborative tools. This provides a highly immersive and interactive learning experience, helping students to develop critical thinking, analytical, and problem-solving skills.

The Bachelor of Computer Applications (BCA) programme is highly appropriate to be conducted in Open and Distance Learning mode as it offers flexibility, self-paced learning, accessibility, technology-driven learning, and personalized learning. These all are essential for students to acquire the skills and competencies required for successful careers in the field of computer science and applications.

E. Instructional Design:-

Total credits of the BCA programme is 145. The Teaching & Examination Scheme is as follows.

Semester-wise distribution of Credits for BCA:

Sr. No.	Semester	Number of Credits
1	Semester – 1	21
2	Semester – 2	20
3	Semester – 3	23
4	Semester – 4	25
5	Semester – 5	26
6	Semester – 6	30
	Total Credits	**145**

- **Teaching & Examination Scheme**

Semester - I										
Course Code	Course Title	Teaching Scheme					Examination Scheme			Total
		Contact Hours				Theory		Practical		
		Theory	Practical	Total	Credit	Internal	External	Internal	External	
OBCA121	Information Technology Trends	3	-	3	3	30	70	-	-	100
OBCA123	Foundation of Programming	4	-	4	4	30	70	-	-	100
OBCA123 P	Foundation of Programming - Practical	-	3	3	3	-	-	30	70	100
OBCA124	Digital Computer Fundamentals	4	-	4	4	30	70	-	-	100
OBCA125	Fundamentals of Web Designing - Practical	-	3	3	3	-	-	30	70	100
OHS101 C	Communicative English	-	4	4	4	-	-	30	70	100
	Total	11	10	21	21	300		300		600

Semester - II											
Course Code	Course Title	Teaching Scheme				Examination Scheme				
		Contact Hours			Credit	Theory		Practical		Total
		Theory	Practical	Total		Internal	External	Internal	External	
OBCA126	Fundamentals of E-Commerce	3	-	3	3	30	70	-	-	100
OBCA128	Advanced Web Designing	-	3	3	3	-	-	30	70	100
OBCA129	Fundamentals of Object Oriented Programming	4	-	4	4	30	70	-	-	100
OBCA129 P	Fundamentals of Object Oriented Programming - Practical	-	3	3	3	-	-	30	70	100
OBCA130	Database Fundamentals	4	-	4	4	30	70	-	-	100
OBCA130 P	Database Fundamentals - Practical	-	3	3	3	-	-	30	70	100
Total		11	09	20	20	300		300		600

Semester - III										
Course Code	Course Title	Teaching Scheme				Examination Scheme				
		Contact Hours			Credit	Theory		Practical		Total
		Theory	Practical	Total		Internal	External	Internal	External	
OBCA227	Computer Oriented Management System	4	-	4	4	30	70	-	-	100
OBCA228	System Analysis and Design	3	-	3	3	30	70	-	-	100
OBCA229	Fundamentals of Data Structures and Algorithms	4	-	4	4	30	70	-	-	100
OBCA229 P	Fundamentals of Data Structures and Algorithms - Practical	-	3	3	3	-	-	30	70	100
OBCA230	Fundamentals of Operating Systems	4	-	4	4	30	70	-	-	100
OBCA230 P	Fundamentals of Operating Systems - Practical	-	3	3	3	-	-	30	70	100
OCL144.01 C	Environmental Sciences	-	2	2	2	-	-	30	70	100
Total		15	8	23	23	400		300		700

Semester - IV										
Course Code	Course Title	Teaching Scheme				Examination Scheme				
		Contact Hours			Credit	Theory		Practical		Total
		Theory	Practical	Total		Internal	External	Internal	External	
OBCA231-232	Elective - IV	-	3	3	3	-	-	30	70	100
OBCA233	Software Engineering	4	-	4	4	30	70	-	-	100
OBCA234	Advanced Database System	-	3	3	3	-	-	30	70	100
OBCA235	Open Source Technology	4	-	4	4	30	70	-	-	100
OBCA235 P	Open Source Technology - Practical	-	3	3	3	-	-	30	70	100
OBCA236	Computer Networks	4	-	4	4	30	70	-	-	100
OHS111.02 C	Human Values and Professional Ethics	-	2	2	2	-	-	30	70	100
OHS121.02 C	Creativity, Problem Solving and Innovation	-	2	2	2	-	-	30	70	100
	Total	12	13	25	25	300		500		800

Elective-IV	
Course Code	**Course Title**
OBCA231	Fundamentals of Embedded Programming
OBCA232	Foundation of Multimedia
SWAYAM courses will be included as electives and updated periodically	

Semester – V										
Course Code	Course Title	Teaching Scheme				Examination Scheme				
		Contact Hours			Credit	Theory		Practical		Total
		Theory	Practical	Total		Internal	External	Internal	External	
OBCA324-325	Elective-V	4	-	4	4	30	70	-	-	100
OBCA324 P – 325 P	Elective-V - Practical	-	3	3	3	-	-	30	70	100
OBCA327	Object Oriented Programming using JAVA	4	-	4	4	30	70	-	-	100
OBCA327 P	Object Oriented Programming using JAVA - Practical	-	3	3	3	-	-	30	70	100

OBCA328	Fundamentals of Visual Programming	4	-	4	4	30	70	-	-	100
OBCA328 P	Fundamentals of Visual Programming - Practical	-	3	3	3	-	-	30	70	100
OBCA329	Introduction to Multi-Paradigms Programming language	-	3	3	3	-	-	30	70	100
OHS131.02C	Communication and Soft skills	-	2	2	2	-	-	30	70	100
Total		12	14	26	26	300		500		800

Elective-V	
Course Code	**Course Title**
OBCA324, OBCA324 P	Document Databases
OBCA325, OBCA325 P	Foundation of Data Science
SWAYAM courses will be included as electives and updated periodically	

Semester - VI										
		Teaching Scheme				Examination Scheme				
		Contact Hours				Theory		Practical		
Course Code	Course Title	Theory	Practical	Total	Credit	Internal	External	Internal	External	Total
OBCA330	Basics of Mobile Applications	4	-	4	4	30	70	-	-	100
OBCA330P	Basics of Mobile Applications - Practical	-	3	3	3	-	-	30	70	100
OBCA331	Software Testing	-	3	3	3	-	-	30	70	100
OBCA332	Minor Project	-	18	18	18	-	-	100	200	300
OHS132.03C	Contributor Personality Development	-	2	2	2	-	-	30	70	100
	Total	04	26	30	30	100		600		700

- **Duration of the programme**

The minimum duration of the programme is 3 years and maximum duration is 5 years.

- **Faculty and support staff requirement**

Category	Existing
Professor	01
Associate Professor	01
Assistant Professor	03

Supporting staff will be deputed at the learner supported centre as per the need of course curriculum.

- **Instructional delivery mechanisms**

As the programme will offer in Online mode, there are various instructional delivery mechanisms and media will be used to effectively deliver content like Learning management system, webinars, video lectures, multimedia, online discussion forum as per requirement. A learning management system (LMS) is an online platform that provides a centralized location for students to access learning content, engage in discussions, submit assignments, and take assessments. The LMS provides a user-friendly interface that is accessible on multiple devices, such as desktops, laptops, tablets, and smartphones. Webinars can be used for lectures, discussions, or interactive sessions with students. Pre-recorded video lectures can be used to deliver course content in a concise and engaging way. Interactive multimedia includes simulations, games, and quizzes that are designed to reinforce learning. Discussion forums can be used to facilitate group discussions, peer-to-peer learning, and to provide feedback and support. Online and face-to-face counselling will be provided by academic counsellors appointed for the programme.

F. Procedure for admissions, curriculum transaction and evaluation:-

The eligibility for the admission is passed in 10+2 examination or equivalent. Students have the convenience of accessing admission information through the university website or contacting the helpdesk number. They can download the admission form from the university website and send it through either online or offline mode. Upon receipt, the university will scrutinize the documents and process the payment of fees. Once the fees are cleared, the admission will be confirmed, and an enrollment number will be issued to the student.

- **Fees Structure**

The programme fee structure will be as follows:

Course	**No. of Semesters**	**Fee for 1st Year**	**Fee for 2nd Year**	**Fee for 3rd Year**	**Total Fee** (for the entire programme)
BCA Programme Fees	6	50,000 INR	50,000 INR	50,000 INR	1,50,000 INR
Course Examination Fee (1500 INR/ Semester)		3000 INR	3000 INR	3000 INR	9000 INR

- **Evaluation System**

To ensure uniform system of education, duration of undergraduate programme, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. **System of Education**
   The Semester system of education should be followed across the Charotar University of Science and Technology (CHARUSAT) at Bachelor's levels. Each semester will be at least 90 working days duration. Every enrolled student will be required to take a specified load of course work in the chosen course of specialization and also complete a project/dissertation if any.

2. **Duration of Programme**

Undergraduate Programme	**Bachelor of Computer Applications(BCA)**
Minimum	6 Semesters (3 Academic years)
Maximum	10 Semesters (5 Academic years)

3. **Eligibility & Mode of Admissions**

Eligibility of a candidate and mode of admission to the programme will be according to the regulations for admission committee decided by Charotar University of Science and Technology from time to time.

4. Programme structure and Credits

A student admitted to a programme should study the course and earn credits specified in the course structure.

5. Course Evaluation

The performance of every student in each course will be evaluated as follows

Internal evaluation by the course faculty member(s) based on continuous assessment, for 50% of the marks for the course; and

Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of these, for 50% of the marks for the course.

University Examination

The final examination by the University for 50% of the evaluation for the course will be through written paper or practical test or oral test or presentation or a combination of these.

In order to earn the credit in a course, a student has to obtain a grade other than F (Fail) or Ab (Absent).

- **Performance at Internal Evaluation Components and University Examination** o If a student secures minimum passing marks of 36% in the University examinations in any course but fails to obtain the minimum passing total percentage of 36%, he/she has to repeat the university examination in the course.

Minimum percentage marks in University Exam for pass in any course	Minimum total percentage marks (i.e. internal + University) for pass in any course
36%	36%

- **Grading**

  o The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

- **Grading Scheme:**

Grading Scheme Range of Marks (%)	96.0-100	86.0-95.9	76.0-85.9	66.0-75.9	56.0-65.9	46.0 – 55.9	36.0 – 45.9	Below 36.0	Absent
Letter Grade	**O** (Outstanding)	**A+** (Excellent)	**A** (Very Good)	**B+** (Good)	**B** (Above Average	**C** (Average)	**P** (Pass)	**F** (Fail)	**Ab** (Absent)
Grade Point	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**0**	**0**

- The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) SGPA = $\sum C_iG_i / \sum C_i$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i

and i = 1 to n, n = number of courses in the semester

(ii) CGPA = $\sum C_iG_i / \sum C_i$ where C_i is the number of credits of course i

G_i is the Grade Point for the course i

and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed.

- No student will be allowed to move to the second academic year if his/her CGPA is less than 3 at the end of the first academic year.
- In addition to above, a student has to comply with the requirements of the regulatory bodies, wherever such requirements exist.
- A student will have a maximum of four chances* after first appearing in that examination to clear that course.

(*Whenever the university conducts the examinations of that course, it will be considered as a chance, irrespective of whether the student appears for the examination or not.)

6. Detention Rule

A student will be promoted to next year only if he/she has cleared all the courses of the year he/she is studying in.

7. Awards of Degree

7.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

7.1.1 He should have earned at least minimum required credits as prescribed in course structure; and

7.1.2 He should have cleared all evaluation components in every course; and

7.2 The student who fails to satisfy minimum requirement of CGPA will be a lowed to improve the grades so as to secure a minimum CGPA for the award of degree. Only latest grade will be considered.

8. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	**CGPA ≥7 & ≤10**
First class:	**CGPA ≥ 6.0 & <7**
Second Class:	**CGPA ≥ 5.0 & <6.0**
Pass Class:	**CGPA <5.0**

9. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

G. Requirement of the laboratory support and Library Resources:-

In an online BCA program, laboratory support will be provided through various means such as virtual labs, cloud-based labs, or remote access to physical labs. Simulations and virtual labs will be used to provide students with a virtual environment in which they can perform practical tasks. In some cases, it may be possible to provide students with remote access to physical labs. Moreover, Instructors will record practical demonstrations and provide students with access to these videos. Students can watch these videos and practice the tasks on their own computers. Instructors will use video conferencing tools to demonstrate practical tasks and answer students' questions.

H. Cost estimate of the programme and the provisions:-

The programme fee structure will be as follows:

Course	No. of Semesters	Fee for 1st Year	Fee for 2nd Year	Fee for 3rd Year	Total Fee (for the entire programme)
BCA Programme Fees	6	50,000 INR	50,000 INR	50,000 INR	1,50,000 INR
Course Examination Fee (1500 INR/ Semester)		3000 INR	3000 INR	3000 INR	9000 INR

I. Quality assurance mechanism and expected programme outcomes:-

The programme structure of online BCA programme is developed under the guidance of the expert committee and Board of Studies and Faculty Board. It is developed as per the guideline of statutory bodies. It is approved by Board of Studies, Faculty Board and Academic Council of the University. Every year the curriculum of the course will be reviewed as per the need of IT Industry and forwarded to the Board of Studies, Faculty Board and Academic Council with suggestions. The changes in the course curriculum as per the needs and requirements from time to time. The University will help the passed-out students in their placement in different industries through the training and placement cell.

The expected programme outcomes are as follows. The Graduate of BCA programme will be able to:

1. Apply knowledge of computing fundamentals, computing specialization, and domain knowledge to provide effective solution in the area of computing.
2. Identify, formulate, and solve complex computing problems reaching substantiated conclusions using fundamental principles of computing sciences, and relevant domain disciplines.
3. Design and evaluate solutions for computing problems, and design and evaluate systems, components, or processes that meet specified needs by considering several societal aspects.
4. Create, select, adapt and apply appropriate techniques, resources, and modern computing tools for any computing activities, with an understanding of the limitations.
5. Understand and commit to professional ethics, responsibilities, and norms of professional computing practice.
6. Recognise the need, and have the ability, to engage in independent learning for continual development as a computing professional.
7. Demonstrate knowledge and understanding of the computing and management principles and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

8. Communicate effectively with the computing community, and with society at large, about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
9. Understand and assess societal and environmental issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice.
10. Function effectively as an individual and as a member or a leader in diverse teams and in multidisciplinary environments.
11. Identify a timely opportunity and using innovation to pursue that opportunity to create value and wealth for the betterment of the individual and society at large.